self attention → parallel processing.
↳ Training b/w fast & scalable.

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Transformers:-

Transformers is a neural net architecture used to handle seq 2 seq tasks
- Machine translation
- text summarization

→ Same here is encoders decorder

→ Instead of lstm, here is self attention.

Impact of transformers.

- Revolution in nlp.
  ↓
  50 years
  ↓ ↳ SOTA.
  NLP.
- Democrating AI.
  ↓
  GPT → Specific fine tuning.
- Multimodal capability
  giving pics to gpt.
- acc of gen AI
- Unification of deep learning.

⇒ Self → Parallel → ~~S~~table.

→ hyperparameter stable & robust

word embedding → Static
Smart contextual " →

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Advantages:-

- Scalability. - Transfer learning.
- Multimodel I/o. - flexible architec
- ecosystem. encoder only Decoder only
hugging face. Bert GPT.
- Integrated AI technique.

Disadvantages:-

- High computation - Data
- Overfitting - Energy consumption.
- Interpretebility (Black box model nobody knows)

Self attention:-

NLP → words → numbers → vectorization.

Self mechanism is such a mechanism which generate smart contextual embeddings from word embedding.

Calculate
↓
apple. launch phone → S.A → Apple. launch. } outputs.
↓
like a function

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They know that a word used in which particular context.

- In 1 line self attention takes static embedding as input & can generate good contextual embedding which are much better in nlp.

Money bank River bank.

if 1 same bank used → static

if changes with meaning → Dynamic

1st principle approach:-

| Money bank grows | river bank flows. |

bank → 0.3 memory + 0.7 bank + 0.1 grows   0.5 river + 0.4 bank + 0.1 flows.

→ General contextual embeddings gives

We have to change it to task specific contextual embedding.

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→ This is b/c there is no learning parameters at all.

Progress:- Words → machine (didn't understand)

↓

embeddings $\xrightarrow{\text{capture}}$ semantic meaning.

a problem occurs that sometimes doesn't capture context like river bank, & money.

→ Then we develop simple self attention model.

general c.e — Task specific c.e.

So for task specific contextual embeddings.

we take:

money | bank | grows.

e_{money} . e_{bank} ↓ e_{grows}.

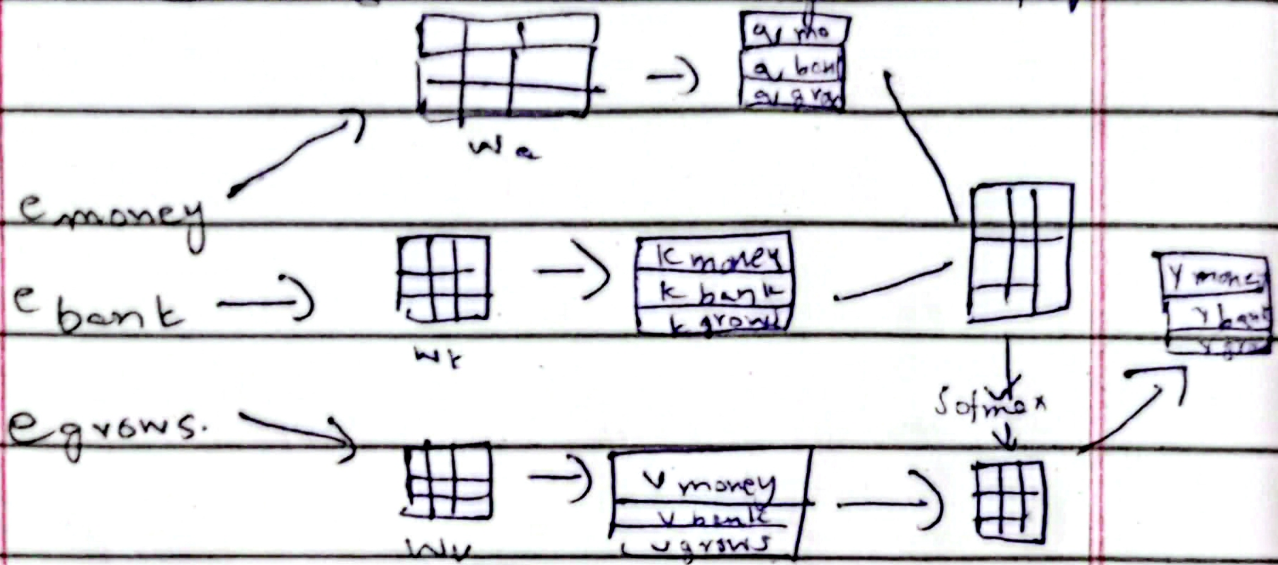
$q_{bank}, k_{bank}, v_{bank}$

$q_{money}, k_{money}, v_{money}$ — $q_{grows}, k_{grows}, v_{grows}$

We divide it to query, key & value

$3 \times 3 = \boxed{9.}$ → new vectors.

We find these 3 vectors from 1 through data for this we do linear transformation & number for matrix we do random initialization & trained through backprop



$$\boxed{\text{Attention}(Q, K, V) = \text{softmax}(QK^T)V} \rightarrow (1)$$

Scaled dot product attention:-

Above formula we derived but in original paper it is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

What is d_k: (Dimentionality of k-vector).

it acts like gravity.
→ means it is context aware & goes to where it needed most.

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When u xply Q & K, from dot product u found a matrix & every value of that matrix u divide by K dimension (means 3 here).

Why:- We do division $\frac{1}{\sqrt{d_k}}$ → b/c dot product ka nature.

- low dimensional → dot product – low variance
- high " → " " – high "

& we will have high dimension vectors → so the variance will be high & that's a problem.

→ for solving that we divide it by $\sqrt{d_k}$.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

& that's used in transformers.

→ but its static

Embedding capture semantic meaning.

means money bank will closer instead of money tree.

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Why its called self attention:-

Turn off the light.

↓ ↓ ↓ ↓

e_1 e_2 e_3 e_4 → embeddings

↓ ↓ ↓ ↓

y_1 y_2 y_3 y_4 → Contextual embed.

q_{turn} k_{turn} v_{turn}

it's intra-sequence not inter-sequence like Luong, & that's why its self.

→ it's attention as methematical formulation is same even way is different.